

Improving GANs using Optimal Transport

A corrected, reproducible and measurable re-implementation

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Originally ENSAE coursework (2021); corrected, modernized and extended (2026)

Abstract

We reproduce the OT-GAN of Salimans et al. [1] on MNIST. Beyond the original implementation, we identify and fix five correctness issues in the training procedure—most importantly, the critic was optimized in the *wrong direction*—and we make the code device-portable (CPU/MPS/CUDA), numerically stable (log-domain Sinkhorn) and quantitatively measurable. Under an upgraded evaluation protocol (FID, KID and a domain-aware FID-LeNet, all at 10,000 samples), the critic-sign fix alone *divides the FID by* ≈ 2.5 ($177.9 \rightarrow 70.8$), turning formless blobs into clearly recognizable digits. We then modernize the objective: replacing the mini-batch energy distance by a *debiased* Sinkhorn divergence [3, 4] removes the documented critic collapse and improves the FID from 81.2 to 24.4 at equal budget, and an optimal-transport conditional flow-matching model [6] built on the *same* Sinkhorn solver reaches FID 11.4. Finally, we show that the same OT engine extends, unchanged, from images to market data: an entropic scenario-reduction tool and a 1D OT-GAN for heavy-tailed returns generation (Sec. VI).

I Generative Modeling and Optimal Transport

Richard Feynman said: “*What I cannot create, I do not understand*”. Generative modeling aims to learn a model of a complex data distribution so as to generate data similar to it. Generative Adversarial Networks (GANs) train a generator G that captures the data distribution and a discriminator D that distinguishes generated from real data; D defines a distance that G minimizes to make its samples resemble the data.

To avoid the tractability problems of the primal optimal-transport formulation, Salimans et al. use mini-batches and a learned cost, defining the *mini-batch energy distance*:

$$D^2(p, g) = 2\mathbb{E}[\mathcal{W}_{c_\eta}(\mathbf{X}, \mathbf{Y})] - \mathbb{E}[\mathcal{W}_{c_\eta}(\mathbf{X}, \mathbf{X}')] - \mathbb{E}[\mathcal{W}_{c_\eta}(\mathbf{Y}, \mathbf{Y}')]$$

where p is the data distribution, g the generator distribution, $\mathbf{X}, \mathbf{X}'$ are independent mini-batches

from p , and $\mathbf{Y}, \mathbf{Y}'$ independent mini-batches from g . The entropic optimal-transport cost is $\mathcal{W}_{c_\eta}(\mathbf{X}, \mathbf{Y}) = \inf_{M \in \mathcal{M}} \text{Tr}[MC^\top]$, with $\mathcal{M}$ the set of non-negative matrices whose rows and columns sum to one, $C_{i,j} = c_\eta(\mathbf{x}_i, \mathbf{y}_j)$, and the learned cosine cost

$$c_\eta(\mathbf{x}, \mathbf{y}) = 1 - \frac{v_\eta(\mathbf{x}) \cdot v_\eta(\mathbf{y})}{\|v_\eta(\mathbf{x})\|_2 \|v_\eta(\mathbf{y})\|_2},$$

where v_η (the *critic*) is a deep neural network. The generator **minimizes** D^2 while the critic v_η **maximizes** it.

II Related Work

Sinkhorn divergences. Entropic regularization makes mini-batch optimal transport tractable on GPU [2]. Genevay, Peyré and Cuturi [3] train generative models with the resulting *Sinkhorn divergence* $S_\varepsilon(p, g) = \mathcal{W}_\varepsilon(p, g) - \frac{1}{2}\mathcal{W}_\varepsilon(p, p) - \frac{1}{2}\mathcal{W}_\varepsilon(g, g)$, whose theoretical properties (positivity, metrization of weak convergence) are established by Feydy et al. [4] for the *debiased* estimator with *same-batch* self-terms $\mathcal{W}(\mathbf{X}, \mathbf{X})$. The mini-batch energy distance of Salimans et al. [1] is exactly *twice* a Sinkhorn divergence whose self-terms are estimated with *independent* mini-batches $\mathcal{W}(\mathbf{X}, \mathbf{X}')$; the distinction between independent- and same-batch self-terms is precisely what our loss study in Sec. V.C isolates.

Mini-batch-OT flow matching. Flow matching [5] trains a velocity field by plain regression along straight noise-to-data paths; Tong et al. [6] pair noise and data through a mini-batch OT plan instead of independently, which straightens the marginal flow and reduces integration error at sampling time. The same Sinkhorn solver that prices mini-batches inside the 2018 GAN can therefore power its 2024 successor. GANs themselves remain competitive when modernized [7].

Evaluation methodology. We follow current practice on metric pitfalls: KID as an unbiased complement to FID [8], aliased-resizing artifacts in FID pipelines [9], the strong bias of FID at small sample counts [10]—the

reason we re-evaluate at 10,000 samples rather than 2,048—and the comparability caveats of the large-scale GAN study of Lucic et al. [11].

III Implementation

We implement the OT-GAN in PyTorch and train it on MNIST ($1 \times 32 \times 32$). The *generator* maps a latent vector to an image: a linear layer followed by a Gated Linear Unit (GLU) reshaped to $512 \times 8 \times 8$, then two (nearest-neighbour upsample, 5×5 convolution, GLU) blocks, and a final convolution with a tanh output. The *critic* maps an image to a 32768-dimensional embedding through three (5×5 convolution, Concatenated-ReLU) blocks, a flatten, and an ℓ_2 normalization—the normalization is what makes the cosine cost well defined. The transport plans are obtained with the Sinkhorn algorithm.

IV Corrections and Improvements

Auditing the original single-notebook implementation revealed five correctness bugs that explain its weak results, and prevented it from running outside CUDA. We fixed all of them (each is covered by a unit test):

1. **Critic trained in the wrong direction (critical).** A single loss was *descended* by both optimizers, so the critic *minimized* the energy distance it must *maximize*. The critic now performs gradient *ascent* ($(-D^2).\text{backward}()$) and the generator gradient descent.
2. **Device portability.** Hard-coded `.cuda()` calls raised on CPU and on Apple-Silicon (MPS). A device resolver (CUDA $\rightarrow$ MPS $\rightarrow$ CPU) makes the code run everywhere; all our experiments run on MPS.
3. **Real/fake range mismatch.** The generator outputs $[-1, 1]$ (tanh) but the real images were in $[0, 1]$, letting the critic cheat on brightness. We normalize images to $[-1, 1]$.
4. **Degenerate second real batch.** `DoubleBatchDataset(batch, batch)` made $\mathbf{X} = \mathbf{X}'$, collapsing $\mathbb{E}[\mathcal{W}(\mathbf{X}, \mathbf{X}')] to ≈ 0 . We now draw two *independent* real mini-batches per step.$
5. **Numerically unstable Sinkhorn.** The raw-exponential iteration $\exp(-C/\varepsilon)$ overflows for small ε . We use a log-domain (`logsumexp`) Sinkhorn; the plan stays detached (envelope theorem), so gradients flow through the cost.

We additionally add a Gaussian latent prior, an exponential moving average (EMA) of the generator used for

sampling, fixed-noise sample grids, and FID/KID/IS evaluation plus a domain-aware FID computed in MNIST-LeNet feature space.

V Experiments and Results

V.A Evaluation protocol

All models in this report are measured with *one* evaluation harness: FID and KID on 10,000 generated versus 10,000 held-out test images (Inception-v3 features), plus a domain-aware *FID-LeNet* computed in the feature space of a LeNet classifier trained on MNIST. The train-vs-test floor of this harness is FID 1.45 / FID-LeNet 1.76. Our earlier ablation numbers were computed on $n = 2048$ evaluation samples; since FID is strongly biased at small n [10], we label those numbers *legacy protocol* and re-evaluate the final checkpoints at 10k. The Inception Score is kept only as a sanity metric: it is essentially identical for the corrected and buggy models (2.33 vs. 2.34), i.e. uninformative on MNIST.

V.B Critic-sign ablation

We train on the full MNIST training set for 18 epochs per run on an Apple-Silicon GPU (MPS), with $\varepsilon = 1$ and 50 Sinkhorn iterations. To isolate the central correction we run an ablation on the critic direction (`critic_sign` on vs. off) at equal budget; quantitative results are reported in Table 1, the per-epoch FID in Fig. 2, and a qualitative comparison in Fig. 1.

critic_sign	legacy (n=2048)	re-evaluation @10k			
	FID final / best	FID	KID $\times 10^3$	FID-LeNet	IS
True (corrected)	77.2 / 69.9	70.8	65.7	77.0	2.33
False (original bug)	185.3 / ~ 170	177.9	191.0	208.8	2.34

Table 1: Critic-sign ablation on MNIST (18 epochs). The left column is the legacy protocol ($n_{\text{eval}} = 2048$, per-epoch); the right block re-evaluates the same final checkpoints at 10k samples. Correcting the critic direction divides the FID by ≈ 2.5 , consistently across all three feature spaces, while the IS does not move.

The corrected model produces clearly recognizable digits, a marked improvement over the original report (whose samples were “recognizable with more or less difficulty”). We also observe that the best FID is reached mid-training (epoch ≈ 8): afterwards the energy distance D^2 collapses toward 0 (a mild critic collapse) and the FID rises slightly, so early stopping on the FID is advisable. The next subsection shows that this collapse is a property of the *objective*, not of the architecture.

Effect of the critic-sign fix at equal budget — MNIST, epoch 8
(in-figure FIDs: legacy in-training protocol, $n_{\text{eval}}=2048$; final-epoch 10k re-evaluations: 70.8 vs 177.9)

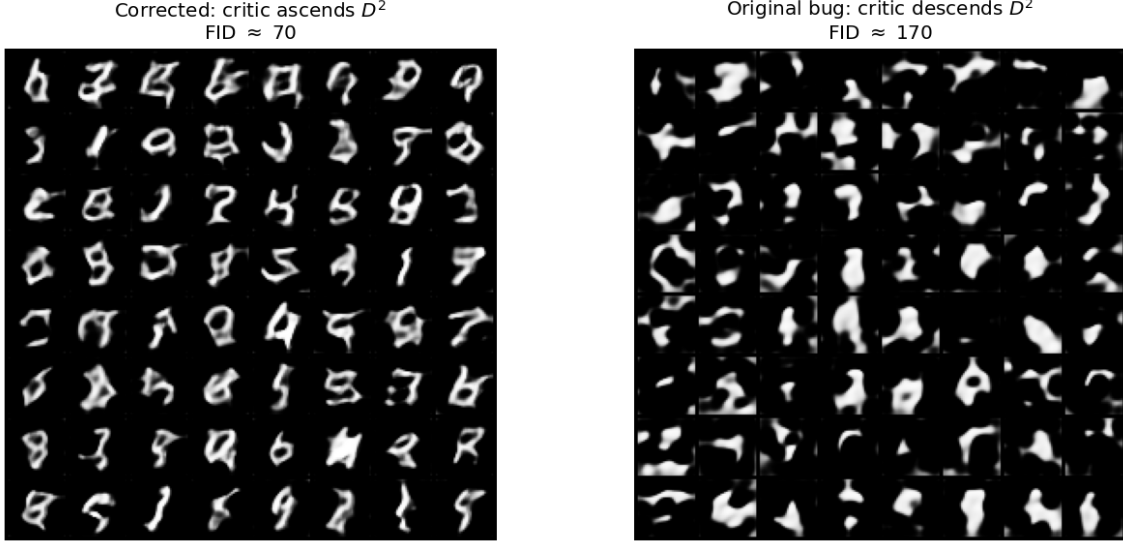


Figure 1: Effect of the critic-sign fix at equal budget (MNIST, epoch 8). *Left*: the corrected model (`critic_sign=True`) produces clearly recognizable digits. *Right*: the original buggy direction (`critic_sign=False`) produces only blobs. The in-figure FID values are the legacy in-training evaluations ($n_{\text{eval}} = 2048$, epoch 8: ≈ 70 vs. ≈ 170); the final-epoch 10k re-evaluations are 70.8 vs. 177.9.

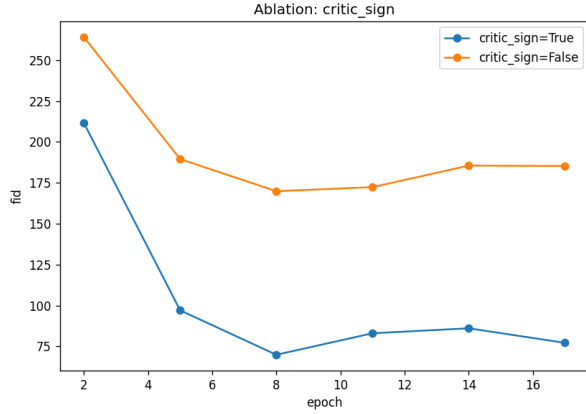


Figure 2: FID vs. epoch (legacy protocol, $n_{\text{eval}} = 2048$). The corrected run (blue) stays well below the buggy run (orange) throughout, reaching its minimum (≈ 70) around epoch 8.

V.C Modernized objective and 2024 successor

As noted in Sec. II, the mini-batch energy distance is exactly twice a Sinkhorn divergence whose self-terms $\mathcal{W}(\mathbf{X}, \mathbf{X}')$ use *independent* mini-batches. We therefore swap in the *debiased* estimator of Feydy et al. [4]—same-batch self-terms $\mathcal{W}(\mathbf{X}, \mathbf{X})$ —keeping everything else identical: same architecture, same seed, same 8-epoch budget. The result (Table 2, Fig. 4) is twofold:

the FID drops from 81.2 to 24.4 ($\text{KID} \times 10^3$: $75.3 \rightarrow 13.8$), and the critic collapse documented above *disappears*—the training objective D^2 stays at ≈ 0.077 instead of collapsing to $\approx 10^{-4}$. The Sinkhorn-divergence run was still improving when the budget ended.

objective (8 epochs, same seed)	FID	KID $\times 10^3$	FID-LeNet	final D^2
energy distance (2018)	81.2	75.3	69.1	1.2×10^{-4}
Sinkhorn divergence (2019)	24.4	13.8	60.1	0.077

Table 2: Equal-budget loss comparison @10k samples. De-biasing the self-terms improves every metric and removes the critic collapse (last column: the objective no longer vanishes).

Finally, we implement the 2024 successor of this line of work: optimal-transport conditional flow matching (OT-CFM) [6, 5], reusing the *same* log-domain Sinkhorn solver to couple noise and data mini-batches—no critic, no minimax, a single regression loss. Table 3 places all models under the single harness of Sec. V.A, together with a DCGAN calibration baseline and an I-CFM control (independent pairs, `cfm_coupling=none`) at identical capacity and budget. The control matches OT-CFM at both 100 and 10 Euler sampling steps (FID 10.9 / 13.3 vs. 11.1 / 13.5 on best checkpoints): at this scale—mini-batch plans over 256 MNIST images—the OT coupling is an honest null result, consistent with its gains concentrating in trajectory straightness, very-low-NFE regimes and

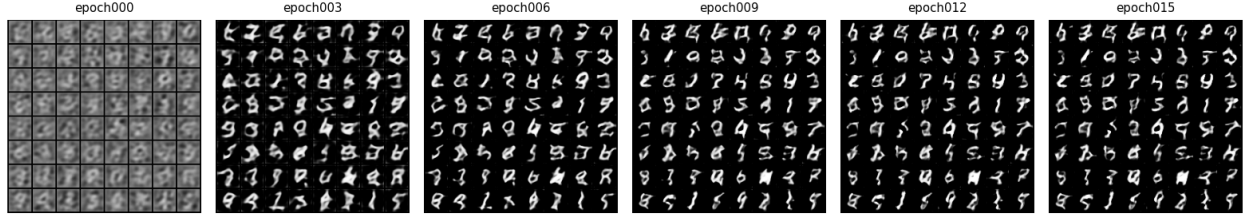


Figure 3: Fixed-noise evolution of the corrected generator: the same latent codes are decoded across epochs, showing the progression from noise to digits.

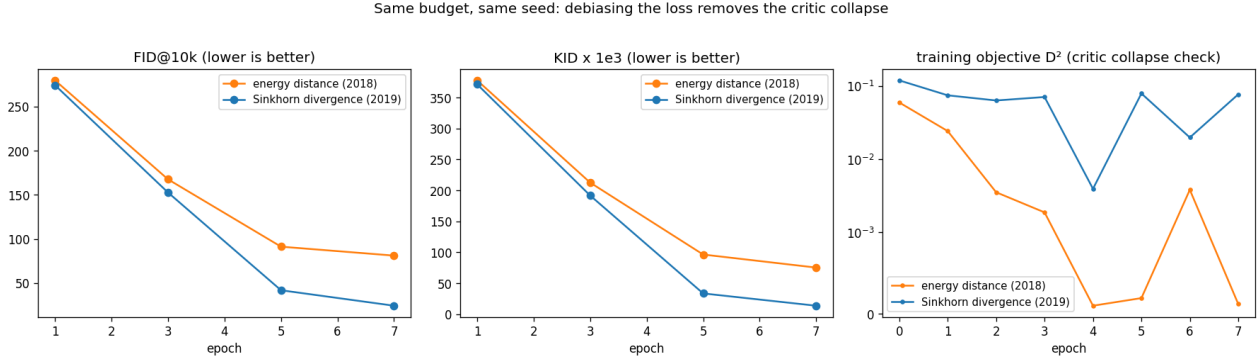


Figure 4: Same budget, same seed: debiasing the loss removes the critic collapse. *Left, middle*: FID@10k and KID $\times 10^3$ per epoch. *Right*: the training objective D^2 (log scale) collapses to $\approx 10^{-4}$ under the 2018 energy distance but stays near 10^{-1} under the debiased Sinkhorn divergence.

harder datasets. Ten-step sampling costs only $\approx +2.7$ FID over the 50-step sweet spot (10.8), though in LeNet feature space ten-step sampling degrades $\approx 2.5\times$ (34.6 vs. 13.6); note also that the GAN generator samples in a single forward pass while the flow pays `ode_steps` forward passes per sample. Repeated evaluations of one checkpoint differ by $\approx \pm 0.3$ FID.

For calibration: Lucic et al. [11] report MNIST FID ≈ 6.7 (WGAN) and ≈ 20.3 (WGAN-GP) at much larger budgets with extensive hyperparameter search. None of our numbers are state-of-the-art claims; they are *controlled comparisons* under one harness at laptop budgets. Note also that FID and FID-LeNet *disagree* on the DCGAN row (72.3 vs. 36.8, against 70.8 vs. 77.0 for the corrected OT-GAN): the domain-aware metric ranks DCGAN clearly better while Inception-FID calls it a tie—itsself evidence that ImageNet-Inception features are unreliable on MNIST.

VI From Images to Markets

Nothing in the OT machinery above is image-specific. We close the experimental part with two finance applications that reuse the *same* log-domain Sinkhorn solver and the *same* training machinery on market data; both ship as executed notebooks under `examples/`, which constitute the full artifacts.

Scenario reduction (no training). Compressing a Monte Carlo book—turning 10,000 simulated paths into K weighted scenarios whose mean, quantiles and CVaR match the full set—is itself an optimal-transport problem. We solve it with an entropic Lloyd iteration (“Sinkhorn k -means”): the plan is computed by the very solver that trains the GAN, centroids are updated by barycentric projection, and as $\varepsilon \rightarrow 0$ the plan hardens to nearest-scenario assignment, recovering classical scenario reduction. The squared-Euclidean cost matrix is normalized ($C \leftarrow C/\text{mean}(C)$) so that ε is *dimensionless* and comparable with the $\varepsilon = 1$ used on cosine costs; reduction runs near the hard limit ($\varepsilon = 0.01$). On a 20,000-path book (10k GBM + 10k Heston paths, 64 daily steps), fitted on one half with all risk statistics evaluated on the held-out half, $K = 100$ scenarios reach a (fit-half) transport distortion of 0.008792, against 0.008866 for k -means and 0.012031 for random subsampling ($\approx 37\%$ worse). Both centroid methods place the terminal mean correctly ($+0.0095$ vs. $+0.0058$ held out) but honestly *understate the tails*: CVaR_{95} 0.080 vs. 0.221 held out, with k -means suffering identically (CVaR fidelity is a tie—OT slightly closer at 95%, k -means at 99%). This contraction is Jensen’s inequality applied to within-cluster averaging, not a solver artifact: the random subsample keeps actual paths, hence unbiased tails (CVaR_{95} 0.163), at the price of representing individual paths $\approx 37\%$ worse—the classical

model	budget (MPS)	FID	KID $\times 10^3$	FID-LeNet	IS
OT-GAN, buggy sign (18 ep)	~ 6 h	177.9	191.0	208.8	2.34
OT-GAN, corrected (18 ep)	~ 6 h	70.8	65.7	77.0	2.33
OT-GAN, energy dist. (8 ep)	~ 3.5 h	81.2	75.3	69.1	2.33
OT-GAN, Sinkhorn div. (8 ep)	~ 3.5 h	24.4	13.8	60.1	1.98
DCGAN baseline (10 ep)	~ 15 min	72.3	74.6	36.8	2.19
I-CFM, no coupling (30 ep)	~ 1.2 h	10.9	8.9	12.8	1.91
OT-CFM, Sinkhorn coupling (30 ep)	~ 1.2 h	11.4	9.6	12.6	1.90
<i>train-vs-test floor</i>	—	1.45	—	1.76	—

Table 3: Headline results, all @10k samples under one harness. Flow matching, built on the same Sinkhorn solver as the GANs, dominates both GAN families in a fraction of the budget; the I-CFM control shows the mini-batch coupling itself is a wash at this scale.

accuracy-versus-tail-fidelity trade-off of scenario reduction.

Heavy-tailed returns generation. A 1D OT-GAN mirrors the image pair component by component—a GLU generator with a *linear* output head (log-returns are unbounded, unlike tanh-bounded pixels) and a dilated-convolution CReLU critic with an ℓ_2 -normalized embedding whose 45-lag receptive field can see volatility clustering—trained with the *same compute_loss* and critic-sign machinery on a seeded GJR-GARCH(1,1) target [12], whose asymmetry $\gamma > 0$ produces a genuine leverage effect that symmetric GARCH lacks. At a 5.5-minute CPU budget (500 of the configured 3,000 steps), every headline stylized fact comes out with the correct *sign* and an overshoot *magnitude* (Table 4); the near-zero long-lag rows (ACF of $|r|$ and r^2 at lags 10–20) fluctuate around zero. A critic-free *debiased* Sinkhorn divergence on raw paths reads 0.0499 against a same-process finite-sample floor of ≈ 0.0096 , i.e. the run is visibly mid-flight. Most tellingly, the critic-sign ablation of Table 1 replays here in minutes: under identical seed and budget, the buggy direction collapses the training objective to +0.0004—it looks “solved”—while its samples carry a lag-1 return autocorrelation of +0.94 (smooth curves, not returns) versus -0.002 for the corrected sign. Calibrating the magnitudes, not just the signs, of temporal dependence is precisely what causal-transport and finance-specific architectures target; this section gestures toward that literature (COT-GAN [13], Quant GANs [14]) with the engine already in place.

VII Limitations

All results are single-seed at laptop budgets (minutes to hours: Apple-Silicon MPS for the image runs, CPU for the finance runs), so the gaps we report are large but not variance-qualified. ImageNet-Inception features are a questionable basis for MNIST evaluation: the IS is identical for corrected and buggy models (2.33 vs. 2.34)

statistic	target	generated
excess kurtosis	+0.9992	+7.0719
ACF(r) lag-1	−0.0123	−0.0022
ACF($ r $) lag-1	+0.0310	+0.2679
ACF(r^2) lag-1	+0.0308	+0.1495
leverage corr lag-1	−0.0243	−0.2926

Table 4: Stylized facts of the GJR-GARCH(1,1) target vs. the 1D OT-GAN at the 5.5-minute CPU budget: right signs (including the leverage asymmetry), overshoot magnitudes—the typical mid-training profile.

and FID/FID-LeNet disagree on the DCGAN ranking, which is why we report three metrics rather than one. The legacy 18-epoch runs kept only final-epoch checkpoints, so their 10k re-evaluations are final-epoch, not best-epoch, numbers. Our per-pair transport cost is the biased primal entropic cost $\text{Tr}[MC^\top]$ (as in the original OT-GAN), not the dual-potential form of Feydy et al., so the per-pair terms are not exactly debiased even when assembled as a Sinkhorn divergence. Finally, the Sinkhorn-divergence run was still improving when its 8-epoch budget ended, so its 24.4 FID understates what the objective can reach.

VIII Code and Reproducibility

The code, configurations, tests and this report are available at <https://github.com/dan-allouche-qf/optimal-transport-gan> (MIT license). Release v1.0.0 ships two weight files with published SHA256 checksums: the EMA generator of the best Sinkhorn-divergence epoch (FID 24.4@10k) and the full OT-CFM checkpoint (FID 11.4@10k); both load through `otgan sample -ckpt`. Each claim maps to one command: `otgan ablate -c configs/mnist.yaml -axis critic_sign -override n_epochs=18` (Table 1), `otgan train -c configs/mnist.yaml -override loss=sinkhorn_divergence` (Table 2), and `otgan train -c configs/cfm_mnist.yaml` (the OT-CFM row of Table 3). The finance track of Sec. VI maps

to the two executed notebooks under `examples/` (its numbers are read off their cell outputs) and to `configs/finance_returns.yaml`. The package is covered by 212 tests (203 fast, 9 slow; $\approx 92\%$ line coverage) run in four CI jobs.

IX Conclusion

A faithful but corrected re-implementation of OT-GAN turns a buggy, CUDA-only notebook into a tested, device-portable package whose corrected critic objective is the decisive factor for sample quality: fixing the critic direction alone divides the FID by ≈ 2.5 on MNIST. Reading the 2018 loss through the Sinkhorn-divergence lens then pays twice—debiasing the self-terms improves the FID from 81 to 24 at equal budget *and* removes the critic collapse—and the same Sinkhorn solver, reused for mini-batch coupling, powers an OT-CFM successor that reaches FID 11.4 in about an hour. Pointed at market data, the identical engine compresses a 20,000-path Monte Carlo book into 100 weighted scenarios with held-out risk fidelity and replays the critic-sign ablation on GJR-GARCH returns in minutes. The full code, configurations, tests and ablation harness are released alongside this report.

References

- [1] T. Salimans, H. Zhang, A. Radford, and D. Metaxas, “Improving GANs using Optimal Transport,” in *ICLR*, 2018.
- [2] M. Cuturi, “Sinkhorn Distances: Lightspeed Computation of Optimal Transport,” in *NeurIPS*, 2013.
- [3] A. Genevay, G. Peyré, and M. Cuturi, “Learning Generative Models with Sinkhorn Divergences,” in *AISTATS*, 2018.
- [4] J. Feydy, T. Séjourné, F.-X. Vialard, S. Amari, A. Trounev, and G. Peyré, “Interpolating between Optimal Transport and MMD using Sinkhorn Divergences,” in *AISTATS*, 2019.
- [5] Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, and M. Le, “Flow Matching for Generative Modeling,” in *ICLR*, 2023.
- [6] A. Tong, K. Fatras, N. Malkin, G. Huguet, Y. Zhang, J. Rector-Brooks, G. Wolf, and Y. Bengio, “Improving and Generalizing Flow-Based Generative Models with Minibatch Optimal Transport,” *TMLR*, 2024.
- [7] Y. Huang, A. Gokaslan, V. Kuleshov, and J. Tompkin, “The GAN is dead; long live the GAN! A Modern GAN Baseline,” in *NeurIPS*, 2024.
- [8] M. Bińkowski, D. J. Sutherland, M. Arbel, and A. Gretton, “Demystifying MMD GANs,” in *ICLR*, 2018.
- [9] G. Parmar, R. Zhang, and J.-Y. Zhu, “On Aliased Resizing and Surprising Subtleties in GAN Evaluation,” in *CVPR*, 2022.
- [10] M. J. Chong and D. Forsyth, “Effectively Unbiased FID and Inception Score and Where to Find Them,” in *CVPR*, 2020.
- [11] M. Lucic, K. Kurach, M. Michalski, S. Gelly, and O. Bousquet, “Are GANs Created Equal? A Large-Scale Study,” in *NeurIPS*, 2018.
- [12] L. R. Glosten, R. Jagannathan, and D. E. Runkle, “On the Relation between the Expected Value and the Volatility of the Nominal Excess Return on Stocks,” *The Journal of Finance*, vol. 48, no. 5, 1993.
- [13] T. Xu, L. K. Wenliang, M. Munn, and B. Acciaio, “COT-GAN: Generating Sequential Data via Causal Optimal Transport,” in *NeurIPS*, 2020.
- [14] M. Wiese, R. Knobloch, R. Korn, and P. Kretschmer, “Quant GANs: Deep Generation of Financial Time Series,” *Quantitative Finance*, vol. 20, no. 9, 2020.